CSC1058 USER INTERFACE DESIGN & IMPLEMENTATION

MODULE COORDINATOR & LECTURER
Hyowon Lee

TEACHING ASSISTANT
Haoyan Zhang

	Tuesday (3 – 4pm)	Wednesday (11am - 12pm)
Week 1 (7 Sep)	Introduction/ orientation	HCI background
Week 2 (14 Sep)	Design guidelines	Design guidelines
Week 3 (21 Sep)	Design guidelines	Design guidelines
Week 4 (28 Sep)	Design guidelines	Project brainstorming
Week 5 (5 Oct)	Guidelines recap	Guidelines recap
Week 6 (12 Oct)	Preventing negative impact	Usability evaluation
Week 7 (19 Oct)	Usability evaluation	Usability evaluation
Week 8 (26 Oct)	Usability evaluation	Usability evaluation
Week 9 (2 Nov)	Usability evaluation	Usability evaluation
Week 10 (9 Nov)	Design principles summary	Visualisation summary
Week 11 (16 Nov)	Project discussion	Project feedback
Week 12 (23 Nov)	Summary/ reflection	Summary/ reflection

1

Guideline 6: Provide short-cuts for frequent use

- Accelerators (ctrl-c/ctrl-v,…)
- Default values / template
- Action history
- Recently-opened
- Infer

Reduces click distance
≈ saves my time

2

Guideline 6:
Provide short-cuts for frequent use

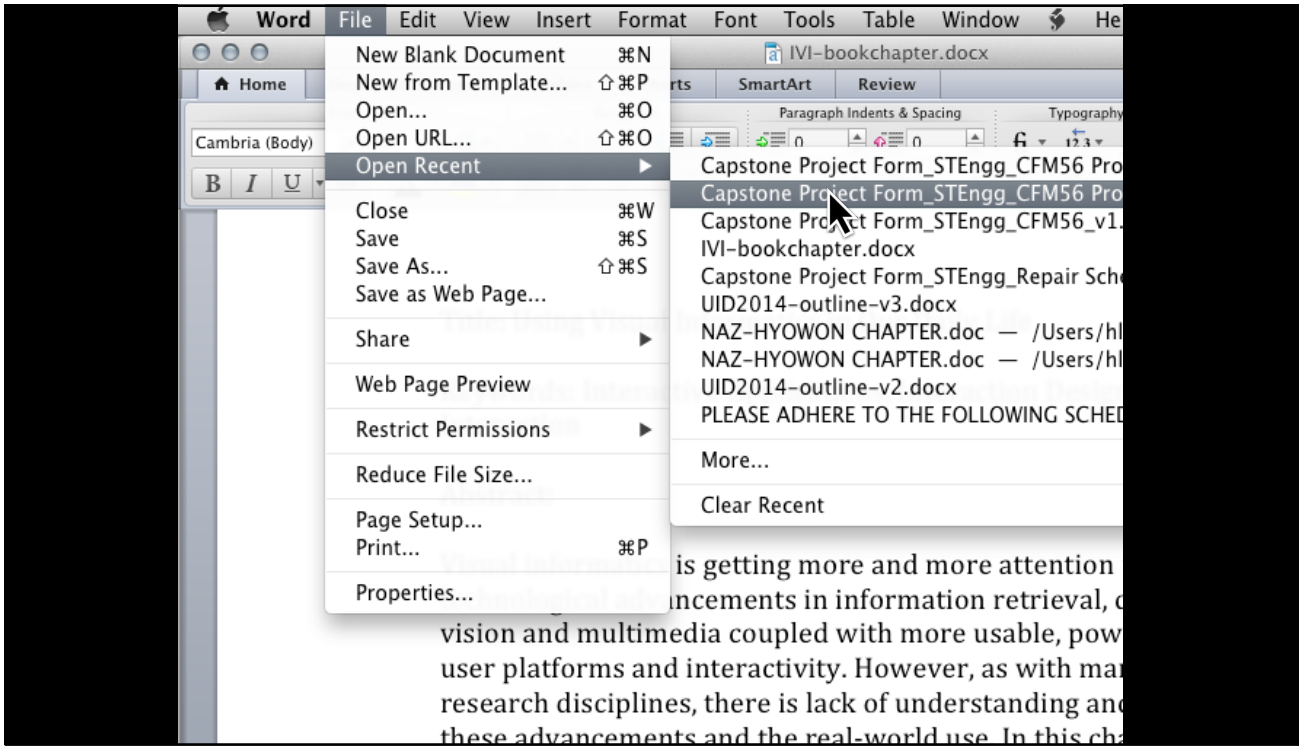
- Accelerators (ctrl-c/ctrl-v,…)
- Default values / template
- tory
- pened

Reduces click distance
≈ saves my time

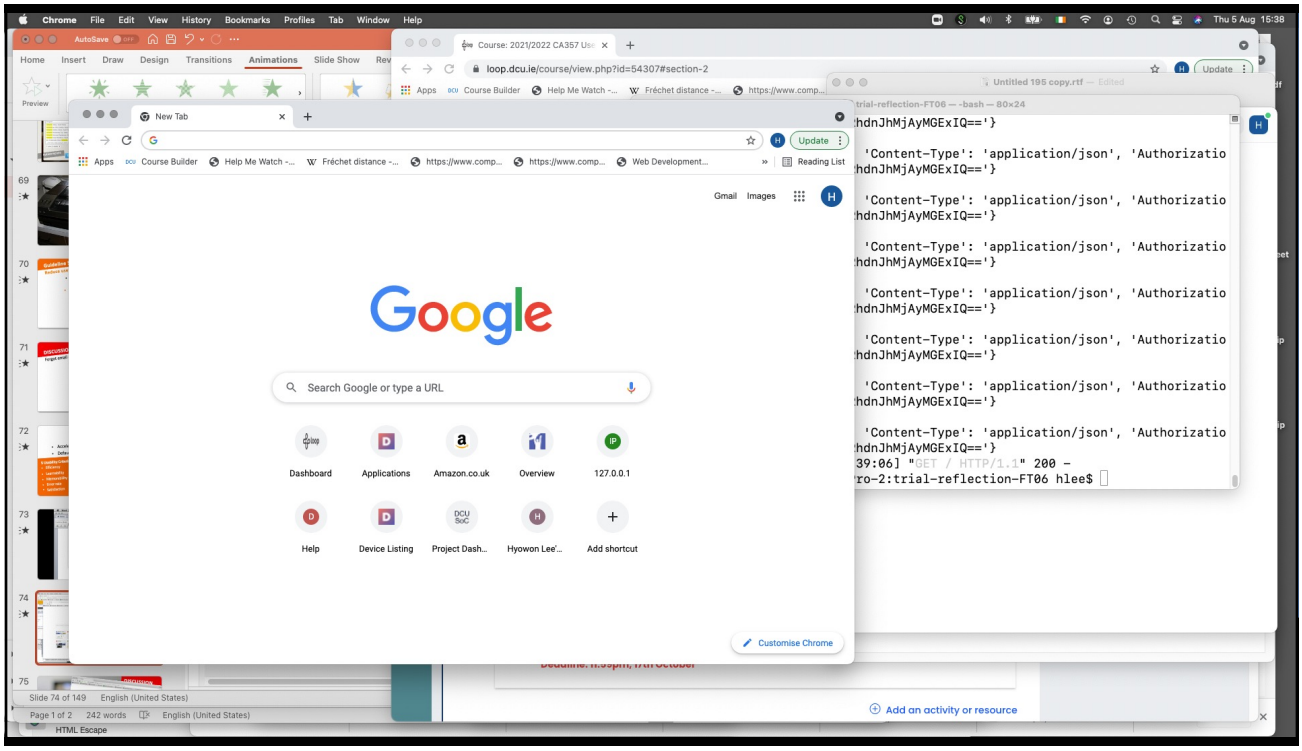
5 Usability Criteria
- Efficiency
- Learnability
- Memorability
- Error rate
- Satisfaction

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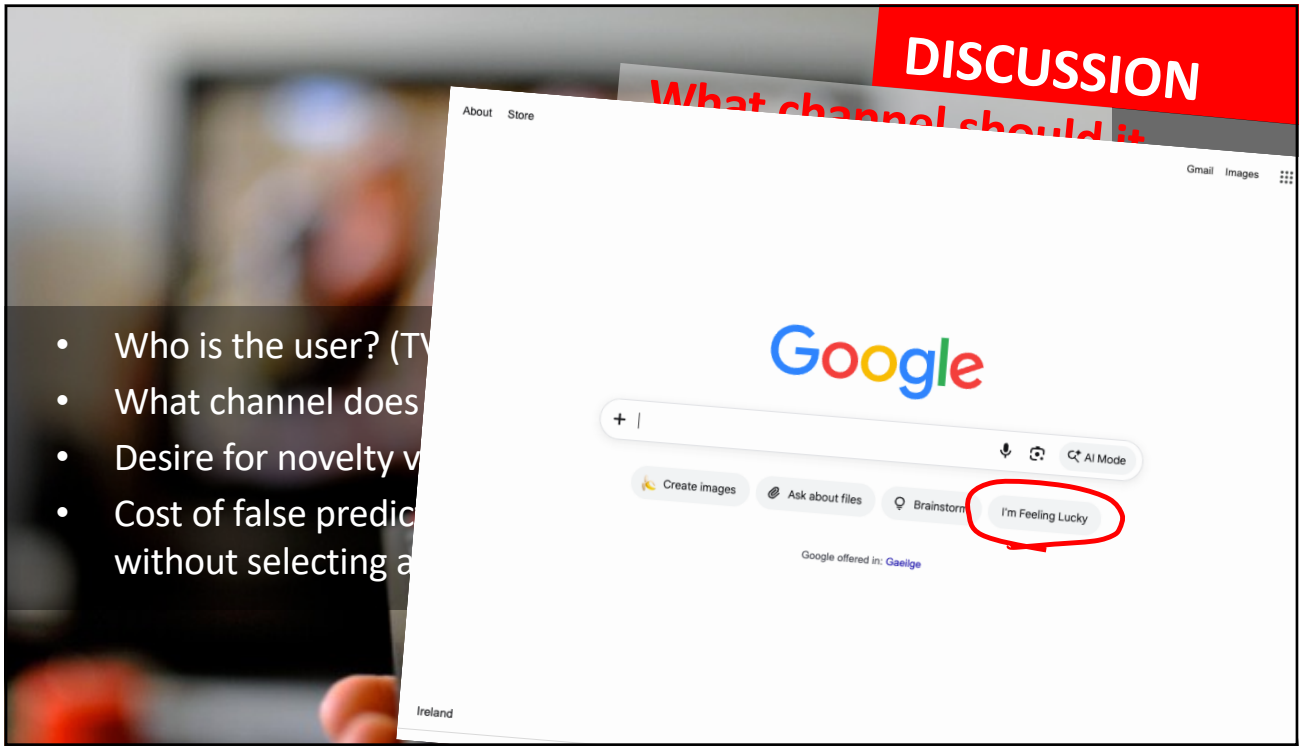
4


Google
Search Google or type a URL
Dashboard
Applications
Amazon.co.uk
Overview
127.0.0.1
Help
Device Listing
Project Dash...
Hyowon Lee'...
Add shortcut
Customise Chrome
Add an activity or resource

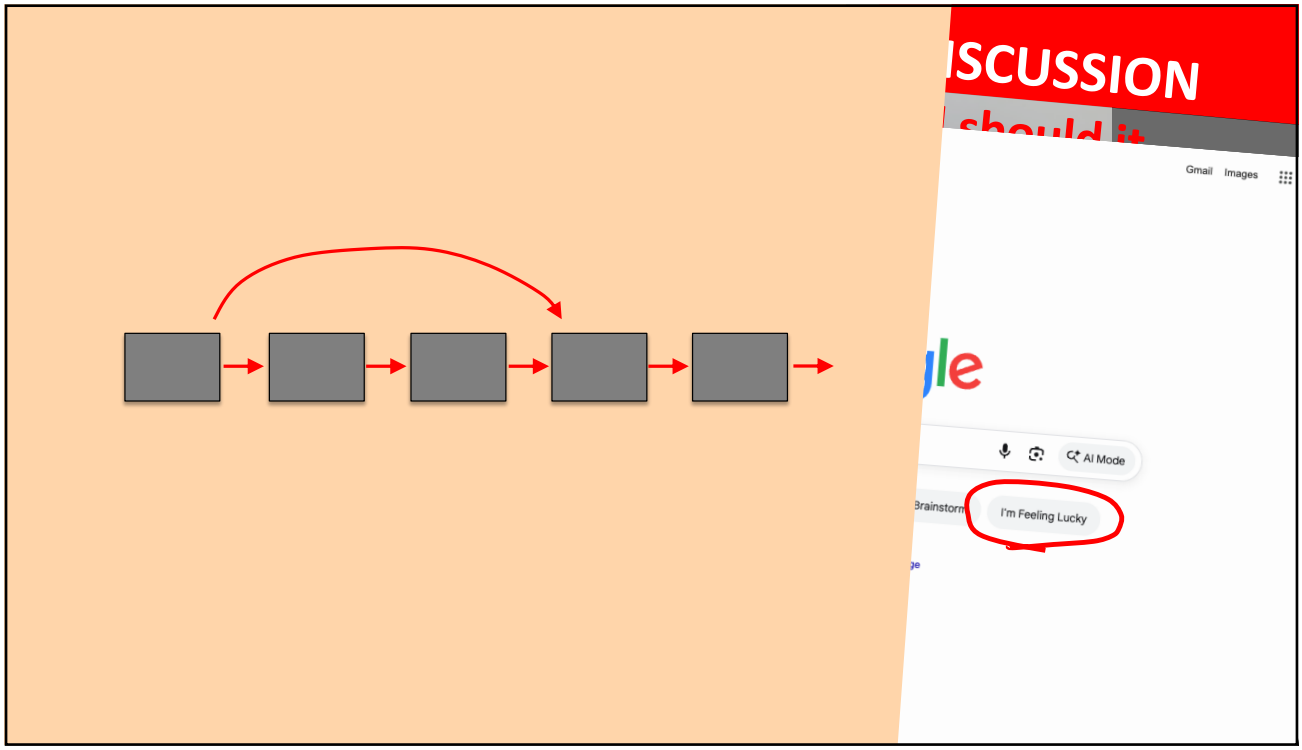
5


DISCUSSION
What channel should it be?... Predicting user needs
Who is the user? (TV vs. smartphone)
What channel does she want to watch?
Desire for novelty vs. desire for high-quality
Cost of false prediction (e.g. online purchasing without selecting an item?)

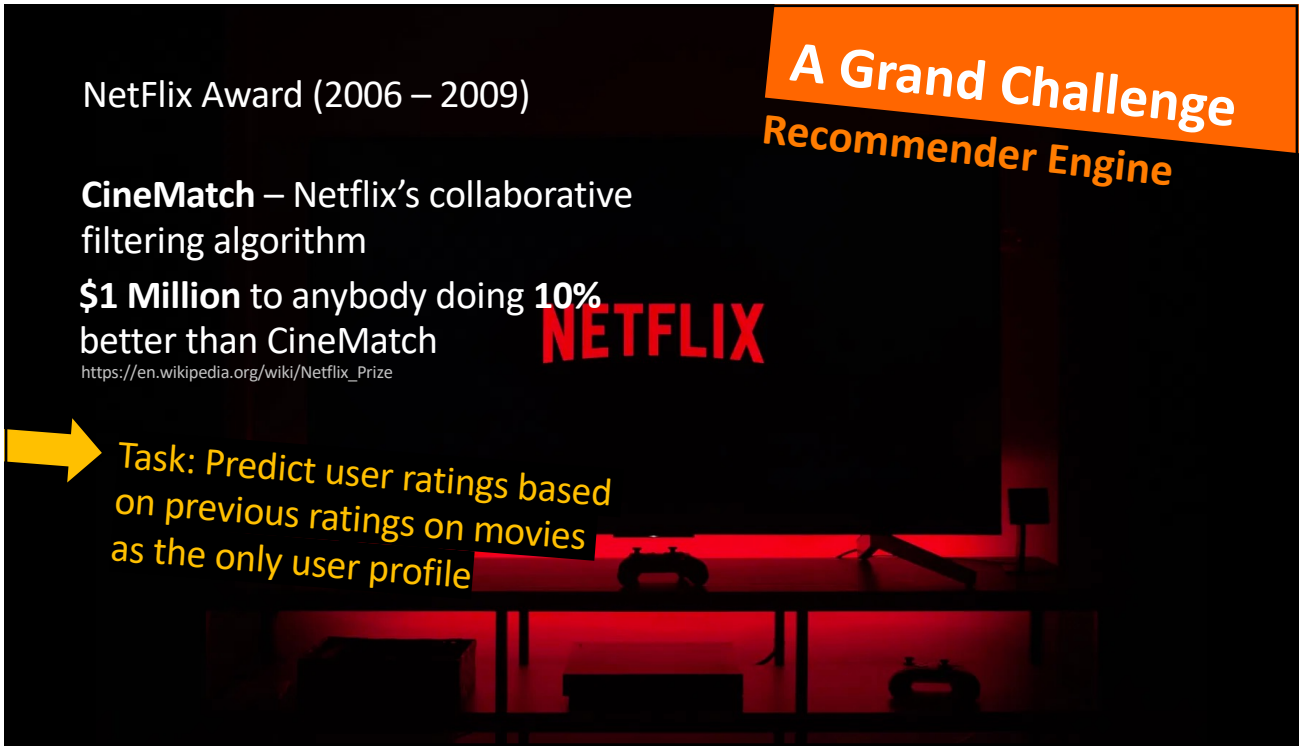
6


DISCUSSION
What channel should it
Who is the user? (TV
What channel does
Desire for novelty v
Cost of false predic
without selecting a
About
Store
Gmail
Images
Google
AI Mode
Create images
Ask about files
Brainstorm
I'm Feeling Lucky
Google offered in: Gaeilge
Ireland

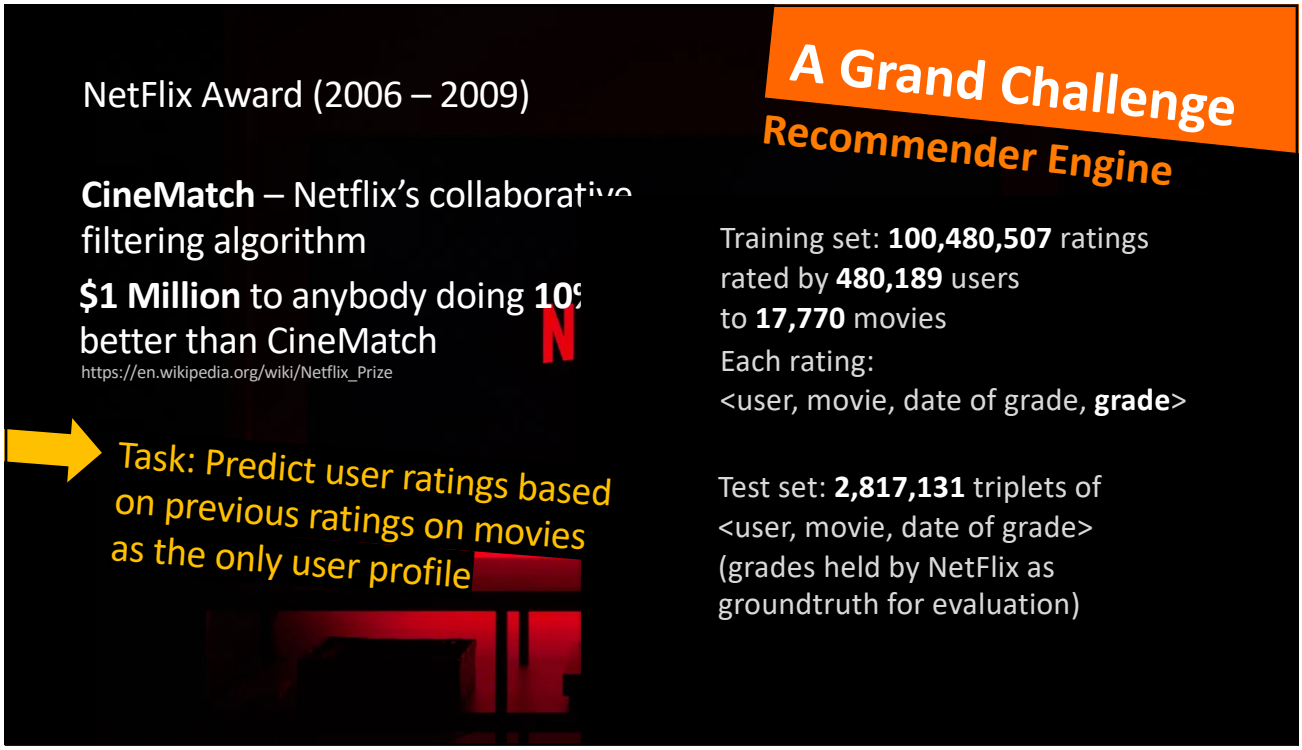
7


ISCUSSION
should it
Gmail
Images
gle
AI Mode
Brainstorm
I'm Feeling Lucky

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A Grand Challenge
Recommender Engine
NetFlix Award (2006 – 2009)
CineMatch – Netflix's collaborative filtering algorithm
$1 Million to anybody doing 10% better than CineMatch
https://en.wikipedia.org/wiki/Netflix_Prize
NETFLIX
Task: Predict user ratings based on previous ratings on movies as the only user profile

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A Grand Challenge
Recommender Engine
NetFlix Award (2006 – 2009)
CineMatch – Netflix's collaborative filtering algorithm
$1 Million to anybody doing 10% better than CineMatch
https://en.wikipedia.org/wiki/Netflix_Prize
Task: Predict user ratings based on previous ratings on movies as the only user profile
Training set: 100,480,507 ratings rated by 480,189 users to 17,770 movies
Each rating:
<user, movie, date of grade, grade>
Test set: 2,817,131 triplets of <user, movie, date of grade> (grades held by NetFlix as groundtruth for evaluation)

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A Grand Challenge

NetFlix Award (2006 – 2009)

CineMatch – Netflix's collaborative filtering algorithm

$1 Million to anybody doing **10%** better than CineMatch
https://en.wikipedia.org/wiki/Netflix_Prize

BellKor's Pragmatic Chaos won (2009)

Bell, R., Koren, Y. and Volinsky, C. Modeling relationships at multiple scales to improve accuracy of large recommender systems. Proceedings of the 13th ACM SIGKDD international conference on Knowledge discovery and data mining, August 2007.



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Guideline 6:

Provide short-cuts for frequent use

- Accelerators (ctrl-c/ctrl-v,…)
- Default values / template
- Action history
- Recently-opened
- Infer

Reduces click distance
≈ saves my time

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DESIGN EXERCISE 2

Re-design chatting UI with "Provide short-cuts for frequent use" in mind (3 people)



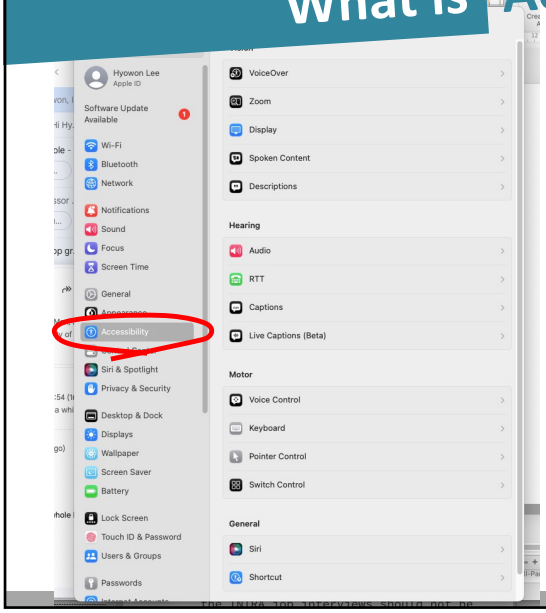
What are frequently-/repeatedly-used actions in a chatting app?

- Default values
- Action history
- Recently-opened
- Reduce click/tap distance
- Predict an action

:

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What is Accessibility?



- Disability issues
- Assistive technology
- Universal design

:

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Accessibility

Making the UI so that users with impairments can use

- Visual impairments
- Hearing impairments
- Motor impairments
- Cognitive impairments

→ *Impairments affect everybody*

- *Aging*
- *Over-exposure (e.g.RSI)*
- *Situational disability*

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Universal Design

Making one UI for everybody

- "Design for All"
- Not sacrificing efficiency
- Often it works, because…

Guiding Principles for UD

- Equitable use
- Flexibility in use
- Simple & intuitive
- Perceptible information
- Tolerance for error
- Low physical effort
- Size and space for approach and use

https://universaldesign.ie/about-universal-design/the-7-principles

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Context Diversity

- Task difference
- Environment difference

One best UI for everybody doesn't exist!

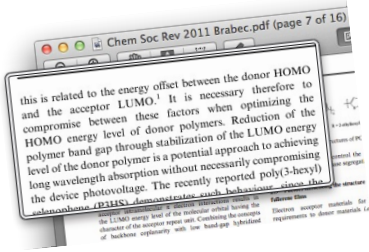
- User difference
  - Learning style
  - Personality traits
  - Motor skill
  - Memory
  - Preference
  - Past experience
  - Culture
  - Language
  - ⋮

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Assistive Technology

Screen magnifier



Screen Reader, Voice Over

Screen flashing on sound

Eye tracker

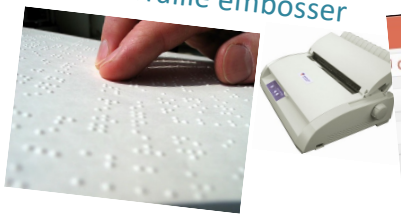
Puf-and-sip device



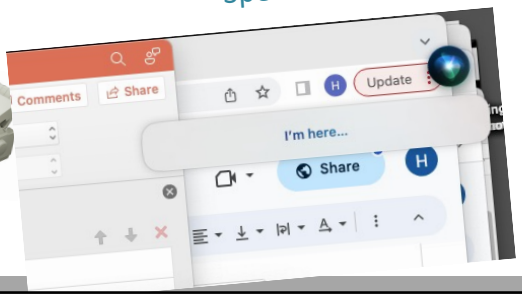
Mouse keys



Braille embosser



Speech recognition





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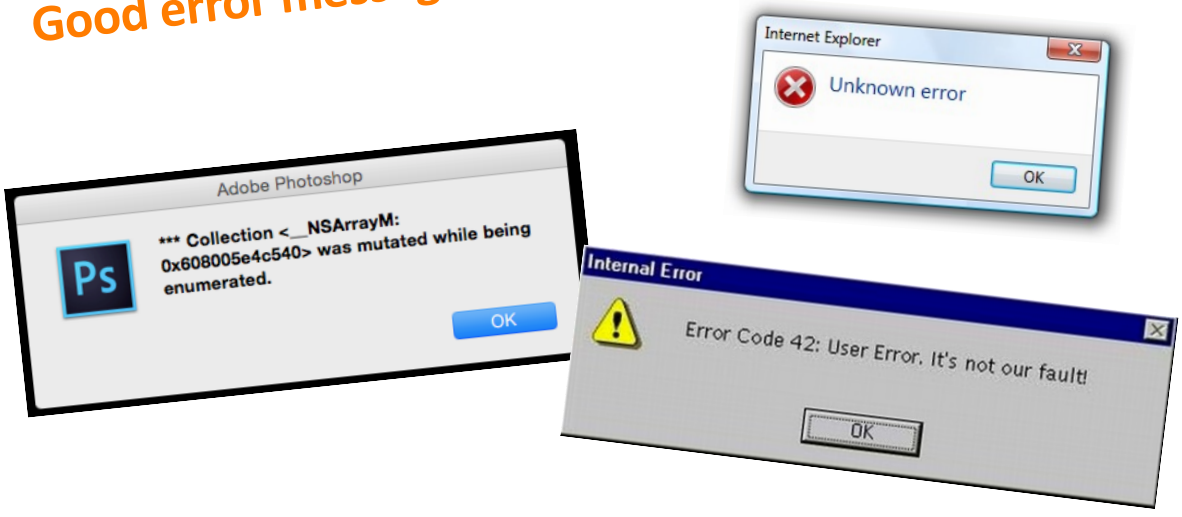
+...

- Customisable font size, background colour, contrast
- Not relying on colour alone
- Flashing as well as beeping
- Closed caption for videos
- Pointing (mouse) interaction having keyboard alternatives

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Guideline 7:
Good error messages



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Guideline 7:
Good error messages

- Four things to remember
  - Be self-explanatory
  - Be specific
  - Be constructive
  - Be polite

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Guid
Goo

"Error 404" → "Page not found"
(be self-explanatory)

"Syntax error" → "Unmatched left parenthesis"
(be specific)

"Undefined labels"
→ "Define statement labels before use"
(be constructive)

"Don't use special characters!!!"
→ "Avoid special characters, instead…"
(be polite)

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Guideline 7: Good error messages

Shred this guideline?

- Four things to remember
  - Be self-explanatory
  - Be specific
  - Be constructive
  - Be polite

If you're designing an error message, think **how error could be avoided** in the first place…

We expect there will be no need to design any error messages in the future (at least for simple consumer applications)

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Guideline 8: Prevent user-driven errors…

- … before designing error messages
- Block any possibility of user-driven error

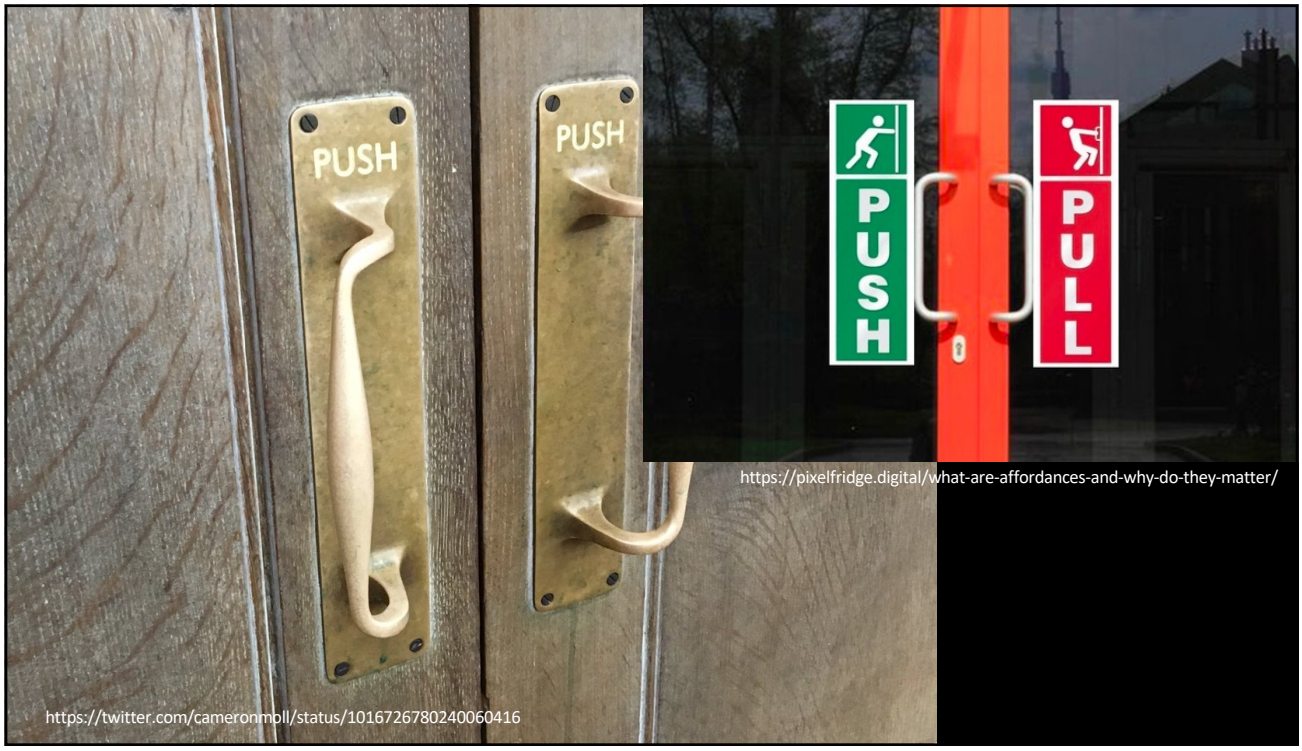
Affordance

Cue or clue that suggests an interaction possibility

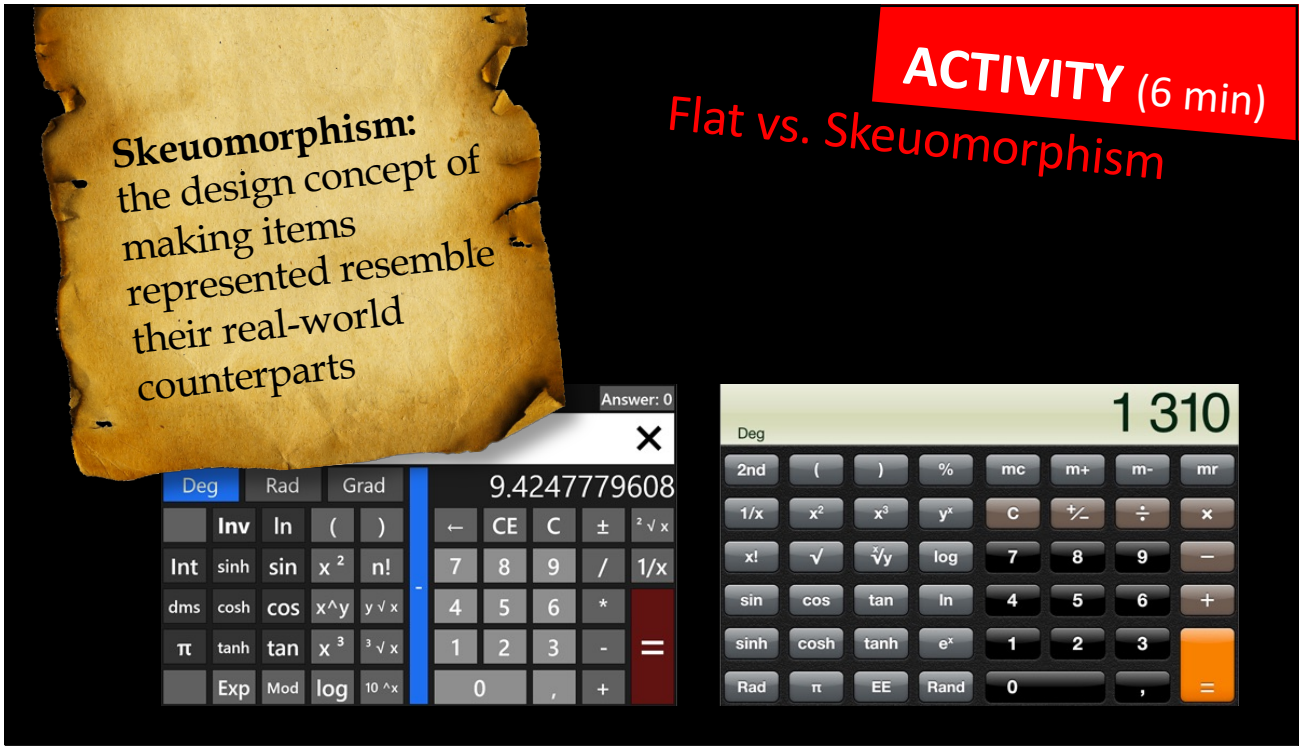
e.g. "a button affords clicking"

"a ball affords grabbing"

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PUSH
PUSH
PUSH
PULL
https://twitter.com/cameronmoll/status/1016726780240060416
https://pixelfridge.digital/what-are-affordances-and-why-do-they-matter/

25


Skeuomorphism:
the design concept of making items represented resemble their real-world counterparts
ACTIVITY (6 min)
Flat vs. Skeuomorphism
Answer: 0
9.4247779608
1 310
Deg
Rad
Grad

26

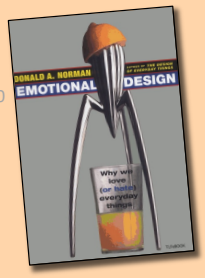

ACTIVITY (6 min)
. Skeuomorphism
Skeuomorphism
Flat
1 310
235
356

27


8:
rs...
Poor affordance
Excellent affordance
... but this doesn't look cool?
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Don Norman, Emotional Design: why we love (or hate) everyday things, Basic Books, 2004.
http://proquestcombo.safaribooksonline.com/9780465004171



3 Levels of Design

- **Visceral** – visual/physical appeal
  - pretty, good-looking, aesthetic, feel good
- **Behavioral** – functional
  - What we normally say 'usability' nowadays
  - Usability engineering
  - "Design by committee"
- **Reflective** – value by the user, image, culture
  - Products can be more than sum of the functions they perform
  - Reflective value can outweigh behavioural difficulty

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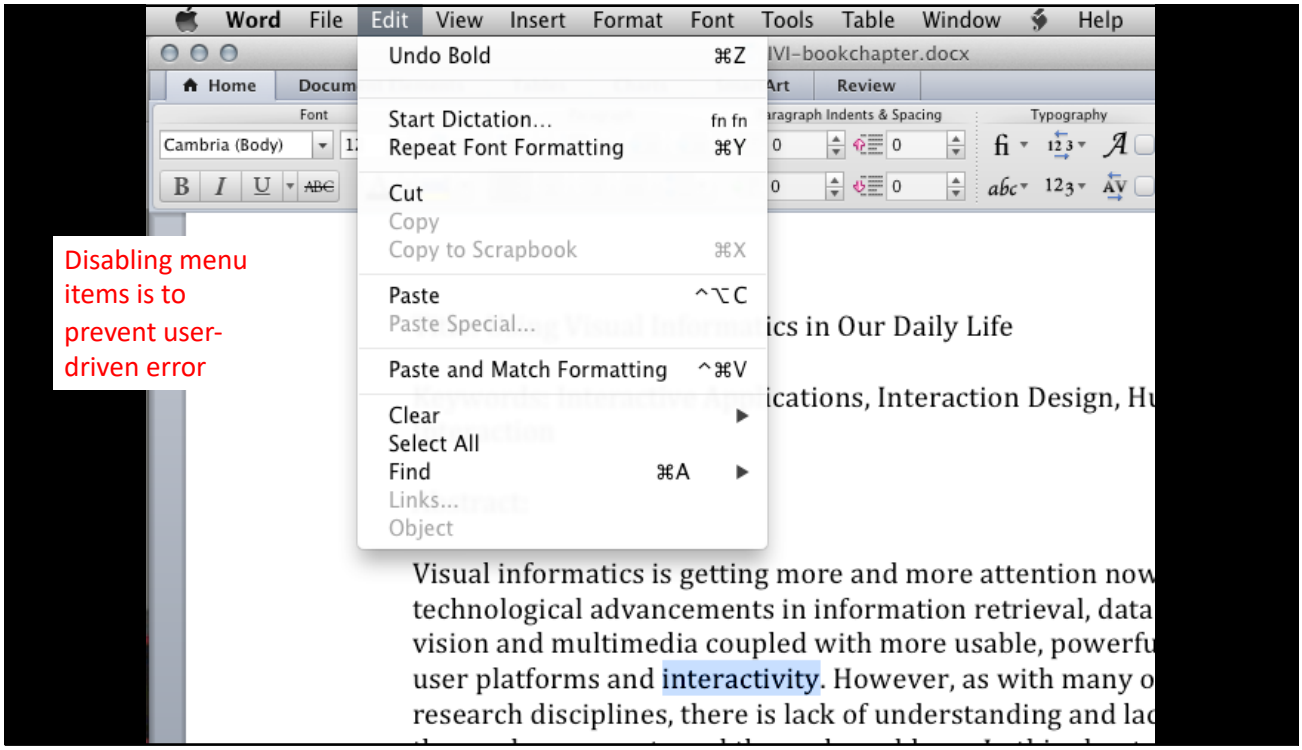
- … before designing error messages
- Block any possibility of user-driven error
- Affordance
- Direct manipulation

- Immediate and continuous feedback
- Incremental and reversable user action
- No user-driven error

➡ Do your best to leave no room for user error

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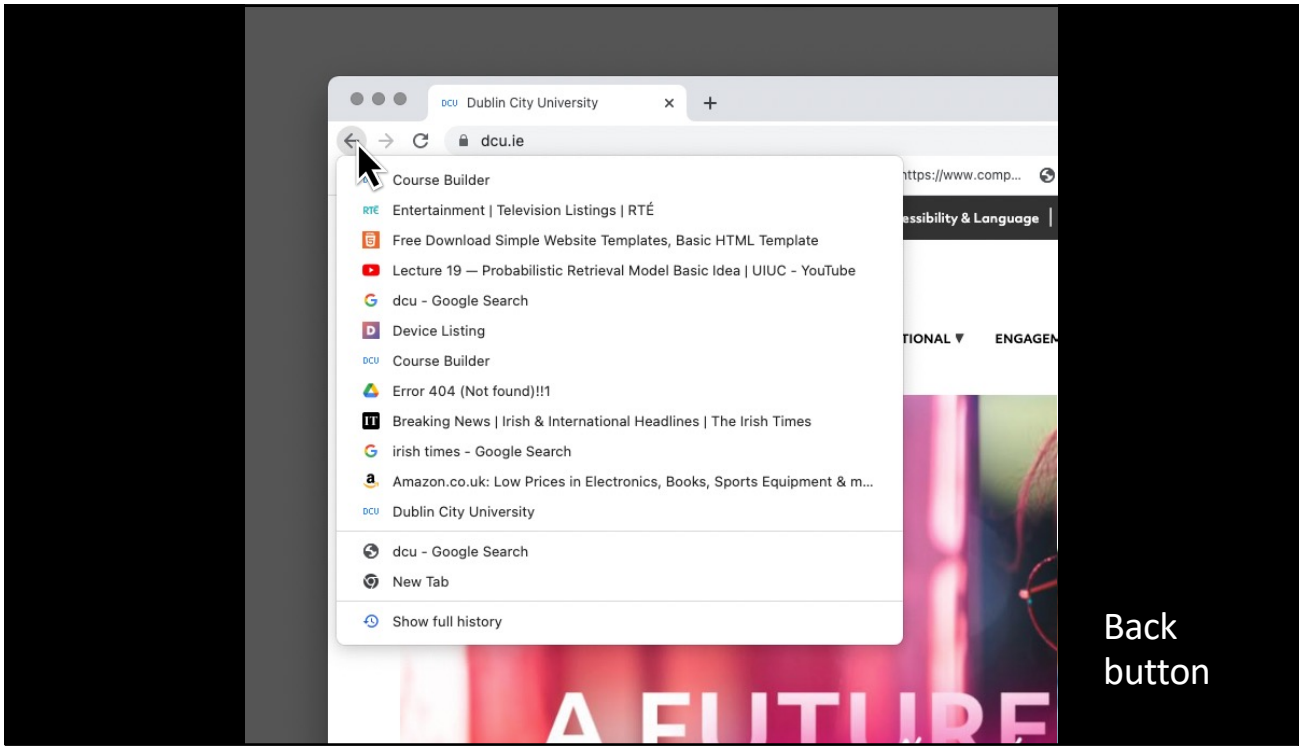


32

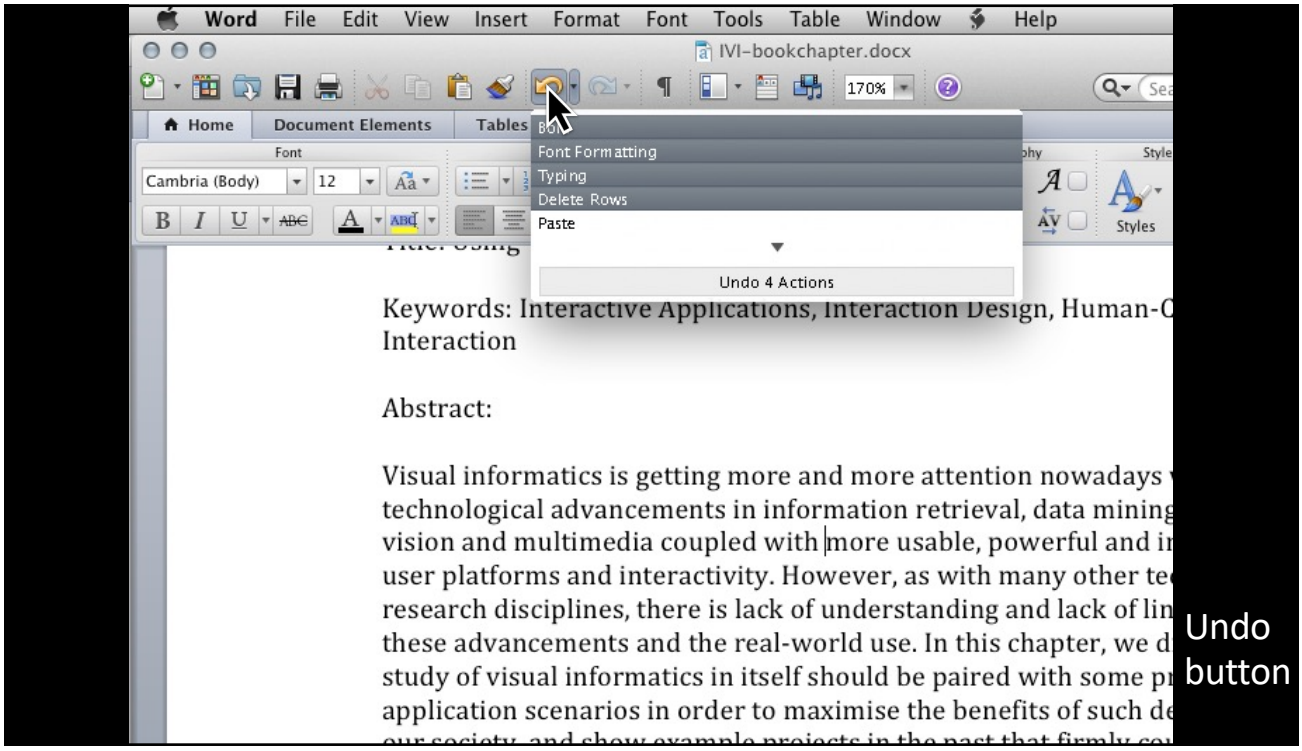
Guideline 9:
Provide easy reversal of action

- Relieves user's anxiety
- Encourages to explore new features, to seek alternatives

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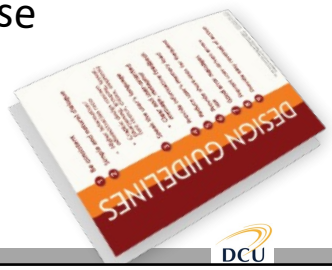
34



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DESIGN GUIDELINE SUMMARY

1. Be consistent
2. Simple and natural dialogue
3. Speak the user's language
4. Provide informative feedback
5. Reduce user's memory load
6. Provide short-cuts for frequent use
7. Good error messages
8. Prevent user-driven errors
9. Provide easy reversal of actions



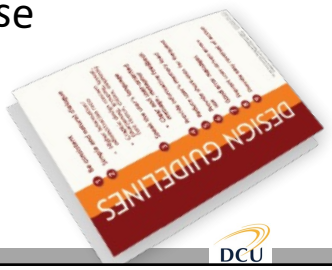
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DESIGN GUIDELINE SUMMARY

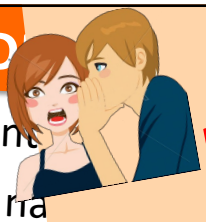
1. Be consistent
2. Simple and natural dialogue
3. Speak the user's language
4. Provide informative feedback
5. Reduce user's memory load
6. Provide short-cuts for frequent use
7. ~~Good error messages~~
8. Prevent user-driven errors
9. Provide easy reversal of actions
10. ~~Help & documentation~~

Maximise info/admin ratio, sharpening, prioritising, grouping, readable fonts, layout, alignment, colour, emphasis



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Hyowon's secret...



I violate the guidelines often...

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DESIGN GUIDELINE SUMMARY

1. Be consistent
2. Simple and natural dialogue
3. Speak the user's language
4. Provide informative feedback
5. Reduce user's memory load
6. Provide short-cuts for frequent use
7. Good error messages
8. Prevent user-driven errors
9. Provide easy reversal of actions

DM is not the result of applying the guidelines, but is explained by the guidelines

Direct Manipulation

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DI

1. Be consistent
2. Simple and na
3. Speak the user
4. Provide informa
5. Reduce user's m
6. Provide short-cu
7. Good error mess
8. Prevent user-drive
9. Provide easy rever

The whole point is…

Use the design guidelines & principles as **conceptual tool** to help you more systematically discuss and argue about the UI, identify alternative UIs that have not been explored

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D

1. Be consistent
2. Simple and na
3. Speak the user
4. Provide informa
5. Reduce user's m
6. Provide short-cu
7. Good error mess
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The whole point is…

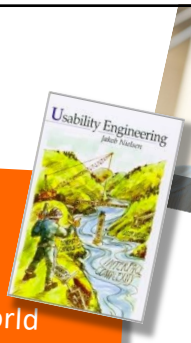
… when you design a UI, focus on the users, not on you

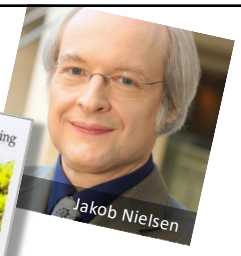
By being careful on what user can do well and not do well, you can design a UI that **compensates user's weaknesses** and **extends their strengths**

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Usability Heuristics

1. Visibility of system status
2. Match between system & world
3. User control and freedom
4. Consistency and standards
5. Error prevention
6. Recognition rather than recall
7. Flexibility & efficiency of use
8. Aesthetic & minimalist design
9. Help error recovery
10. Help and documentation







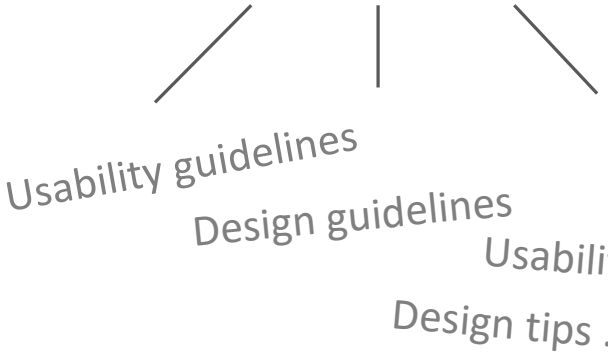


Eight Golden Rules

1. Strive for consistency
2. Cater to universal usability
3. Offer informative feedback
4. Design dialogs to yield closure
5. Prevent errors
6. Permit easy reversal of actions
7. Support internal locus of control
8. Reduce short-term memory load

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Design Principles



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Design Principles

Affordance
Consistency
Constraints
Feedback
Mapping
Visibility

- Derived from a mix of theory-based knowledge, experience, and common sense
- Intended to help designers explain and improve the design

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Design Principles

- Affordance
- Consistency
- Constraints
- Feedback
- Mapping
- Visibility



Designing the User Interface: Strategies for Effective Human-Computer Interaction (6th Ed.) Ben Shneiderman, et al., 2016; Addison-Wesley

Interaction Design: Beyond Human-Computer Interaction (5th Ed.) Jennifer Preece, Yvonne Rogers, Helen Sharp, 2019; John Wiley & Sons

The Design of Everyday Things Don Norman, 2013; Basic Books

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Design Principles

- ▶ **Affordance**
- Consistency
- Constraints
- Feedback
- Mapping
- Visibility

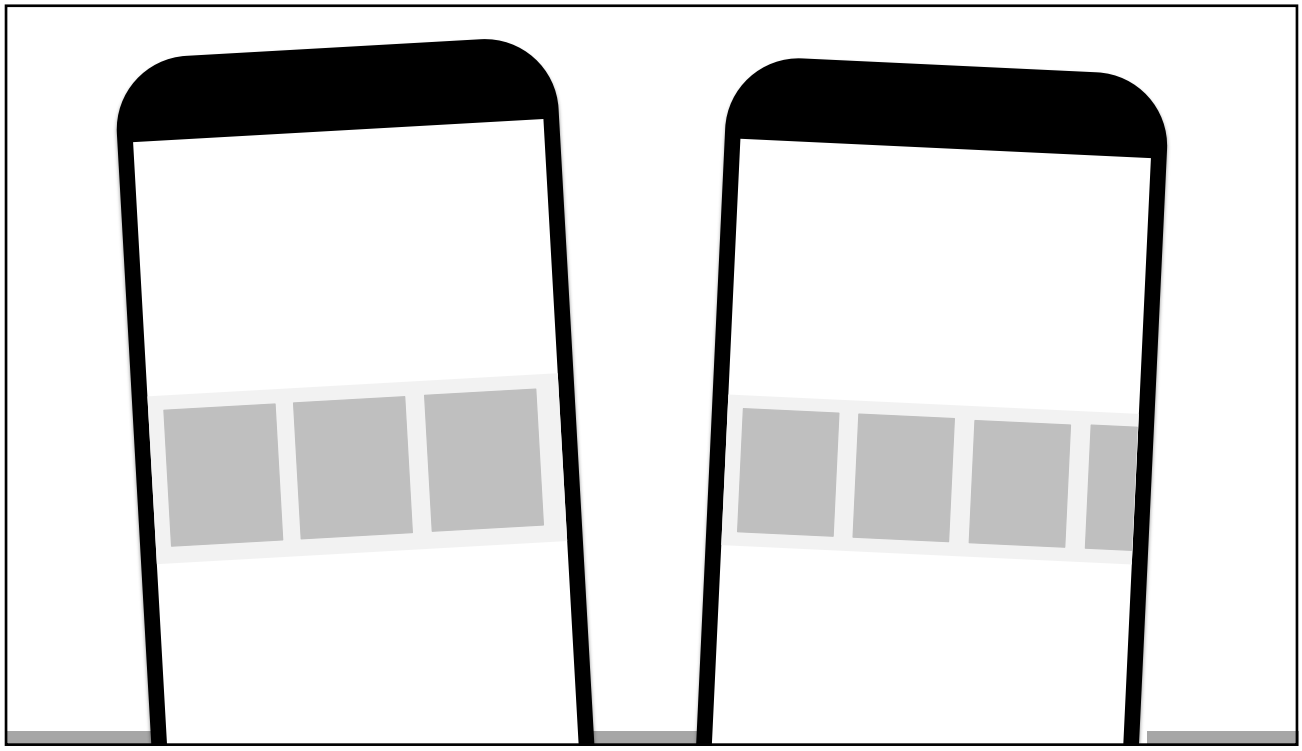
Cue/clue/hint that suggests a particular interaction possibility

e.g. "a button **affords** clicking"

"a ball **affords** grabbing"

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ACTIVITY (8 min)

Missing affordance?

1. Pair with a person beside you
2. Choose two of our day-to-day apps/software you use often (WhatsApp, Word, Netflix, YouTube, Spotify…)
3. For each app/software, identify a **missing affordance** (i.e. where the hints/clues to a particular interaction possibility are absent)
4. Add the affordance by sketching/drawing

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Design Principles

Affordance

▶ **Consistency** .....

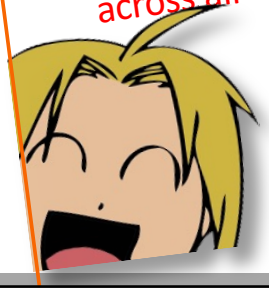
Constraints

Feedback

Mapping

Visibility

→ Layout, colour, font, capitalisation, menu, message style, tone of voice, animation effects, icon style… EVERYTHING Within an environment/screen, across all



→ **Learnability** ("easy to guess how to use")

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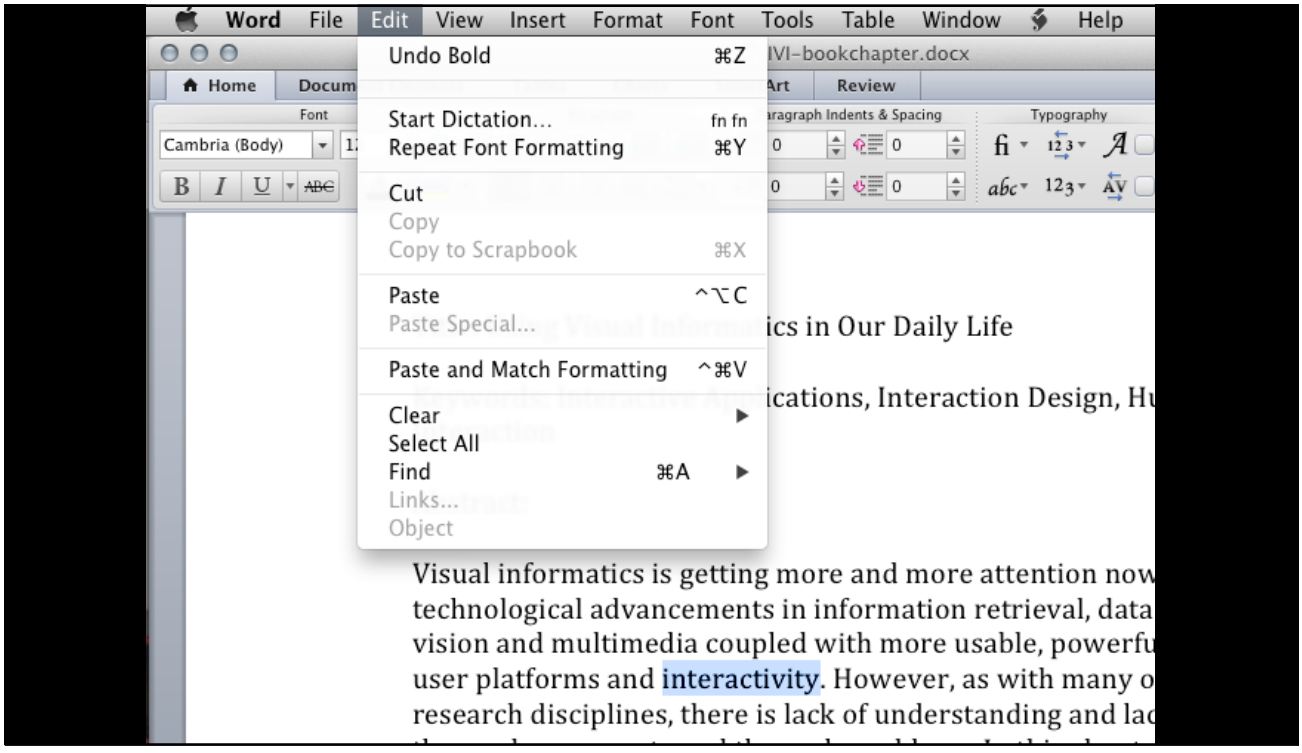
50

Design Principles

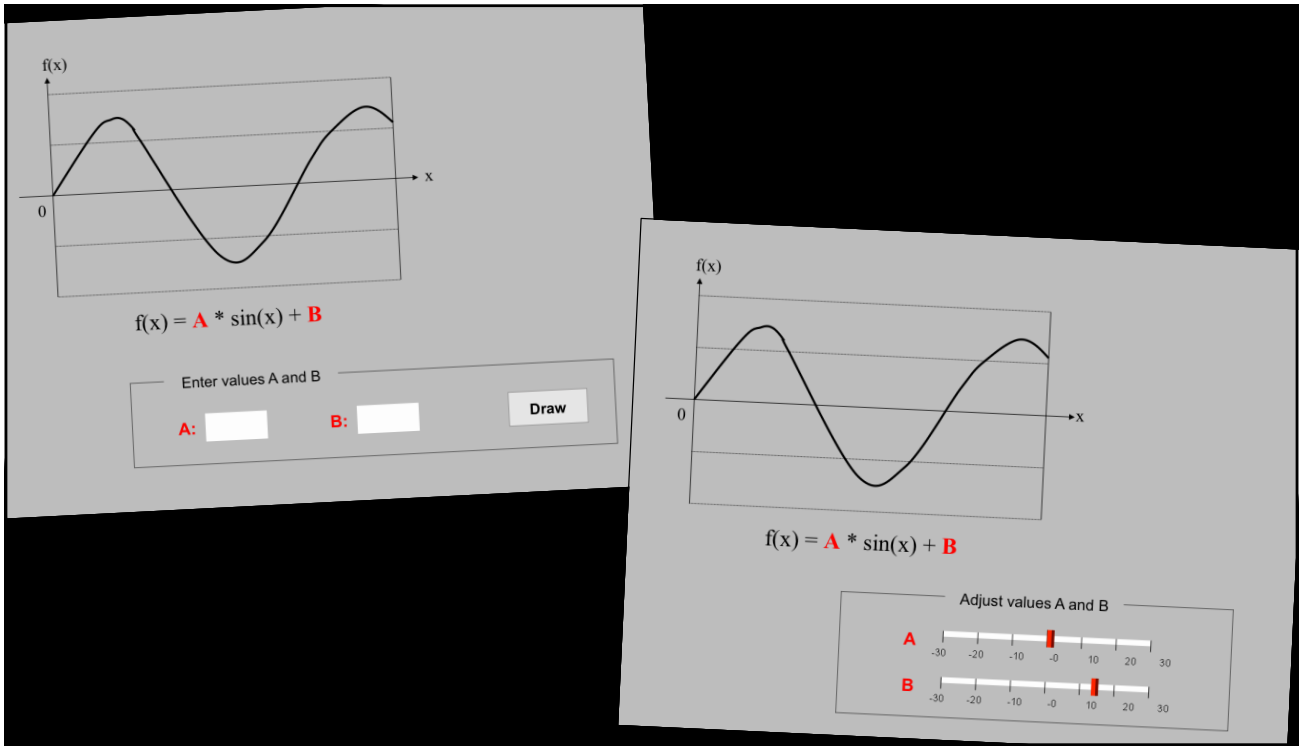
- Affordance
- Consistency
- **Constraints** ..... To prevent user-driven errors… (e.g. Easter annual leave)
- Feedback
- Mapping
- Visibility

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f(x)
x
0
f(x) = A * sin(x) + B
Enter values A and B
A:
B:
Draw
f(x)
x
0
f(x) = A * sin(x) + B
Adjust values A and B
A
B
-30
-20
-10
-0
10
20
30

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